MEDIUM · PROBLEM-SOLVING AND DATA ANALYSIS · ANSWER KEY

Problem-Solving and Data Analysis

03

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

A

A. None

Choice A is correct. Statement I need not be true. The fact that 78% of the 1,000 adults who were surveyed responded that they were satisfied with the air quality in the city does not mean that the exact same percentage of all adults in the city will be satisfied with the air quality in the city. Statement II need not be true because random samples, even when they are of the same size, are not necessarily identical with regard to percentages of people in them who have a certain opinion. Statement III need not be true for the same reason that statement II need not be true: results from different samples can vary. The variation may be even bigger for this sample since it would be selected from a different city. Therefore, none of the statements must be true.

Choices B, C, and D are incorrect because none of the statements must be true.

Q2

B

B. 13

Choice B is correct. It's given that the value of z is 1.13 times 100. This can be written as $z = 1.13100$, which is equivalent to $z = 1 + 0.13(100)$, or $z = 1 + \frac{13}{100}100$. It follows that the value of z is 100% of 100 plus 13% of 100. Therefore, the value of z is 13% greater than 100.

Choice A is incorrect. This gives a value of z that is 1.113, not 1.13, times 100.

Choice C is incorrect. This gives a value of z that is 2.30, not 1.13, times 100.

Choice D is incorrect. This gives a value of z that is 3.13, not 1.13, times 100.

Q3

A

A. $\frac{1}{7}$

Choice A is correct. It's given that there are 2 hybrid cars priced at no more than $25,000. It's also given that there are 14 cars total for sale. Therefore, the probability of selecting a hybrid priced at no more than $25,000 when one car is chosen at random is $\frac{2}{14} = \frac{1}{7}$.

Choice B is incorrect. This is the probability of selecting a hybrid car priced greater than $25,000 when choosing one car at random. Choice C is incorrect. This is the probability, when choosing randomly from only the hybrid cars, of selecting one priced at no more than $25,000. Choice D is incorrect. This is the probability of selecting a hybrid car when selecting at random from only the cars priced greater than $25,000.

Q4

D

D. 2

Choice D is correct. A line in the *xy*-plane that passes through the points x_1, y_1 and x_2, y_2 has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points $1, 3.3$ and $7, 14.5$. It follows that the slope of this best fit line is approximately $\frac{14.5 - 3.3}{7 - 1}$, which is equivalent to $\frac{11.2}{6}$, or approximately 1.87. Therefore, of the given choices, 2 is closest to the slope of the line of best fit shown.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q5

C

C. 2, 4, 8, 16, 32, 64, 128, 256, 512

Choice C is correct. If the values in a data set are ordered from least to greatest, the median of the data set will be the middle value. Since each data set in the choices is ordered and contains exactly 9 data values, the 5th value in each is the median. It follows that the median of the data set in choice C is 32. The sum of the positive differences between 32 and each of the values that are less than 32 is significantly smaller than the sum of the positive differences between 32 and each of the values that are greater than 32. If 32 were the mean, these sums would have been equal to each other. Therefore, the mean of this data set must be greater than 32. This can also be confirmed by calculating the mean as the sum of the values divided by the number of values in the data set: $\frac{2+4+8+16+32+64+128+256+512}{9} = 113\frac{5}{9}$.

Choices A and B are incorrect. Each of the data sets in these choices is symmetric with respect to its median, so the mean and the median for each of these choices are equivalent. Choice D is incorrect. The median of this data set is 207. Since the sum of the positive differences between 207 and each of the values less than 207 is greater than the sum of the positive differences between 207 and each value greater than 207 in this data set, the mean must be less than the median.

Q6

2432

The correct answer is 2,432. It's given that $4 \text{ cups} = 1 \text{ quart}$. It follows that 76 quarts is equivalent to $76 \text{ quarts}\left(\frac{4 \text{ cups}}{1 \text{ quart}}\right)$, or 304 cups. It's also given that $8 \text{ fluid ounces} = 1 \text{ cup}$. It follows that 304 cups is equivalent to $304 \text{ cups}\left(\frac{8 \text{ fluid ounces}}{1 \text{ cup}}\right)$, or 2,432 fluid ounces.

Q7

B

B. Between 28.59% and 35.41% of all adults who own televisions use their televisions to watch nature shows.

Choice B is correct. It's given that based on a sample selected at random, it's estimated that 32% of all adults who own televisions use their televisions to watch nature shows, with an associated margin of error of 3.41%. Subtracting the margin of error from the estimate and adding the margin of error to the estimate gives an interval of plausible values for the true percentage of adults who own televisions who use their televisions to watch nature shows. This means it's plausible that between 32% - 3.41%, or 28.59%, and $32\% + 3.41\%$, or 35.41%, of all adults who own televisions use their televisions to watch nature shows. Therefore, of the given choices, the most plausible conclusion is that between 28.59% and 35.41% of all adults who own televisions use their televisions to watch nature shows.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect. To make a plausible conclusion about all adults who own televisions, the sample must be selected at random from all adults who own televisions, not just those who use their televisions to watch nature shows.

Choice D is incorrect. Since the sample was selected at random from all adults who own televisions, a plausible conclusion can be made about all adults who own televisions.

Q8

A

A. $1.09x$

Choice A is correct. Increasing the positive quantity x by 9% is the result of adding 9% of x to x. 9% of x can be represented algebraically as $\frac{9}{100}x$, or $0.09x$. Adding this expression to x yields $x+0.09x$, or $1.09x$.

Choice B is incorrect. This represents 9% of x. Choice C is incorrect. This represents increasing x by 9, not by 9%. Choice D is incorrect. This represents increasing x by 0.09, not by 9%.

Q9

B

B. $\frac{1}{10}$

Choice B is correct. If 1,200 customers register for new accounts every day, then $(1{,}200)(5) = 6{,}000$ customers registered for new accounts in the first 5 days. Therefore, of the first 60,000 new accounts that were registered, $\frac{6{,}000}{60{,}000}$, or $\frac{1}{10}$, were registered in the first 5 days.

Choice A is incorrect. The fraction $\frac{1}{5}$ represents the fraction of accounts registered in 1 of the first 5 days. Choice C is incorrect and may result from conceptual or computation errors. Choice D is incorrect. The fraction $\frac{1}{50}$ represents the fraction of the first 60,000 accounts that were registered in 1 day.

Q10

6

The correct answer is 6. The line of best fit predicts a greater y-value than the actual y-value for any data point that's located below the line of best fit. For the scatterplot shown, 6 of the data points are below the line of best fit. Therefore, the line of best fit predicts a greater y-value than the actual y-value for 6 of the data points.

Notes

Accept equivalent forms (e.g. `0.25` ≡ 1/4). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank